Verification matrix

The project is large, so "CI is green" says little without a breakdown: below is which claim is proven by what.

Claim	Check	Command	CI
PCAP/PCAPNG parsing neither loses nor invents bytes	`tests/capture/*`	`make test-core`	`ci.yml` → `test`
Stream reassembly survives gaps, overlaps and retransmissions	`tests/capture/test_reassembly*`, `test_fragmentation`, `test_boundaries`	`make test-core`	`ci.yml` → `test`
Every byte is linked to the packet that produced it (provenance)	`tests/capture/test_provenance.py`	`make test-core`	`ci.yml` → `test`
Interpretation rules are deterministic and data-independent	`tests/protocol/test_engine.py`, `test_rule.py`, `test_framing.py`	`make test-core`	`ci.yml` → `test`
Hypotheses and counterexamples are distinguishable (observation ≠ conclusion)	`tests/hypothesis/*`	`make test-core`	`ci.yml` → `test`
Work on the real corpus	`tests/integration/*`, `tests/hypothesis/test_real_corpus.py`	`make test-integration`	`ci.yml` → `test`
The report and the export are reproducible	`tests/report/*`, `tests/integration/test_reference_report.py`	`make test-integration`	`ci.yml` → `test`
The project container is portable	`tests/project/*`	`make test-integration`	`ci.yml` → `test`
The interface does not lose state and is geometrically correct	`tests/ui/*` (offscreen)	`make test-ui`	`ci.yml` → `ui-tests`
The web interface works	`tests/web/*`	`make test-web`	`ci.yml` → `test`
Style and static analysis	`ruff check .`	`make lint`	`ci.yml` → `code-quality`
No known vulnerabilities in the dependencies	`pip-audit`	—	`ci.yml` → `audit`
The built wheel installs and starts	`make package`	`make package`	`portability.yml` → `package`
The documentation is complete, the links are alive and the languages match	`make docs-check`	`make docs-check`	`portability.yml` → `docs`
The environment is suitable before start-up	`make doctor`	`make doctor`	locally

What is proven locally and what only in CI

Level	Where it is checked
The Qt-free core (`make test-core`, 446 tests)	locally in ~15 s and in CI
The web interface (`make test-web`)	locally and in CI
The interface (`make test-ui`)	only where PySide6 and the system Qt libraries exist
The package (`make package`)	locally in a venv and in CI

The local core tests run without PySide6; `dpkt` and `PyYAML` are mandatory.

What is deliberately not checked

- Parity of the English and Russian documentation sets — checked by `tools/docs_pdf.py --check` since 2026-10-30 (a document without a translation fails `make docs-check`).
- Compatibility with Windows/macOS: the project targets Linux x86-64.
- Performance on captures far larger than the corpus.
- The quality of hypotheses: what is checked is consistency and the presence of counterexamples, not the "truth" of the protocol.